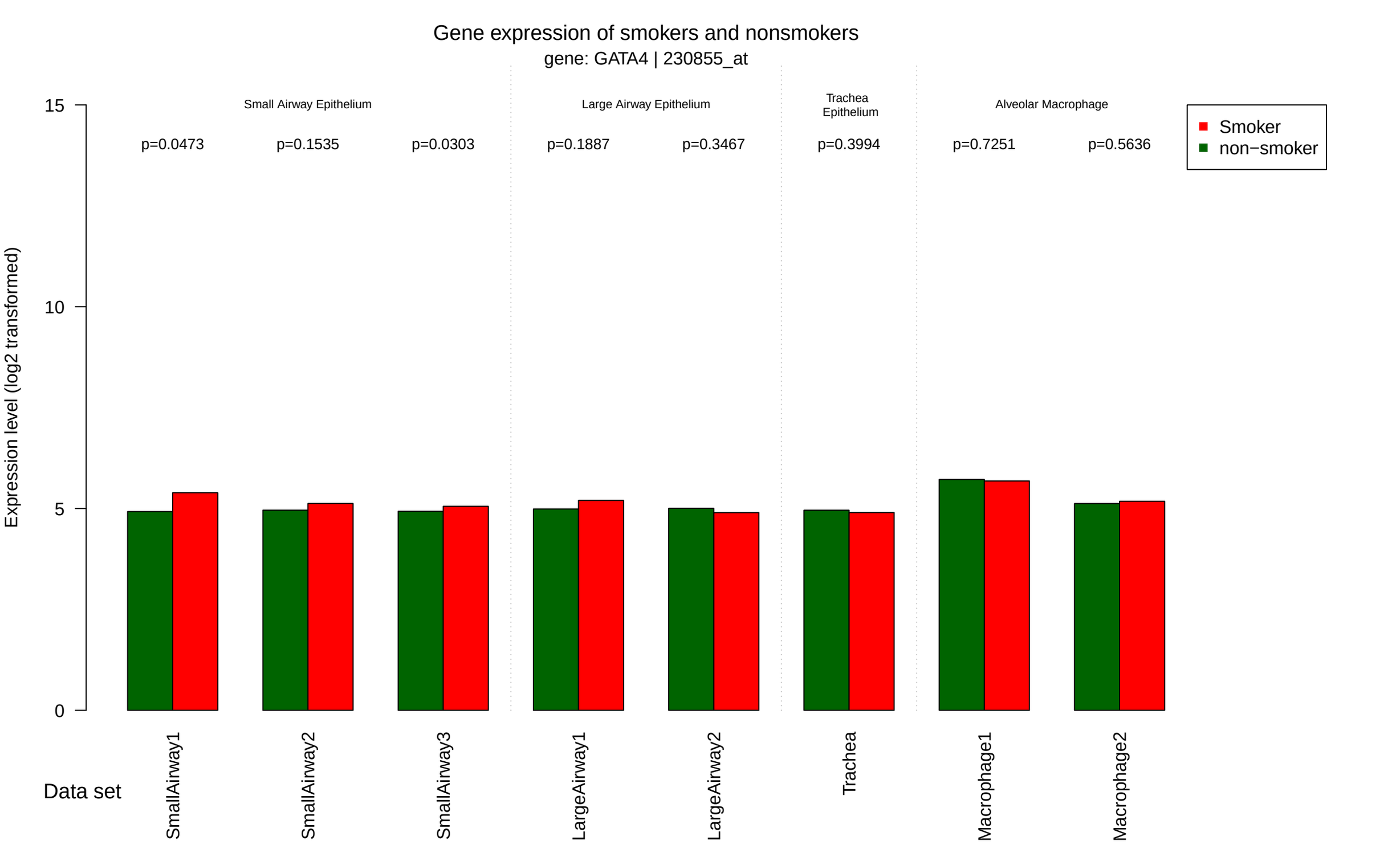
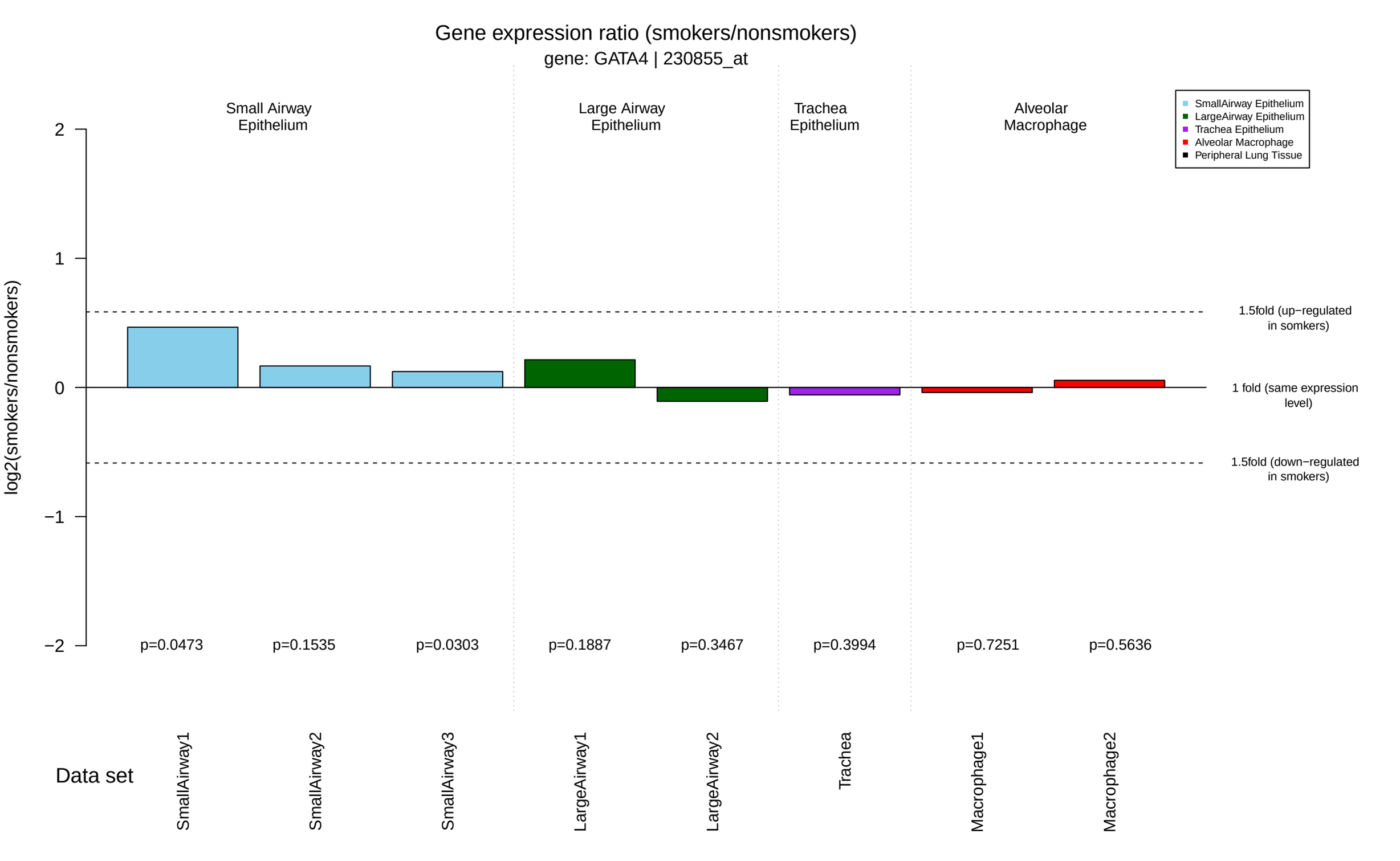
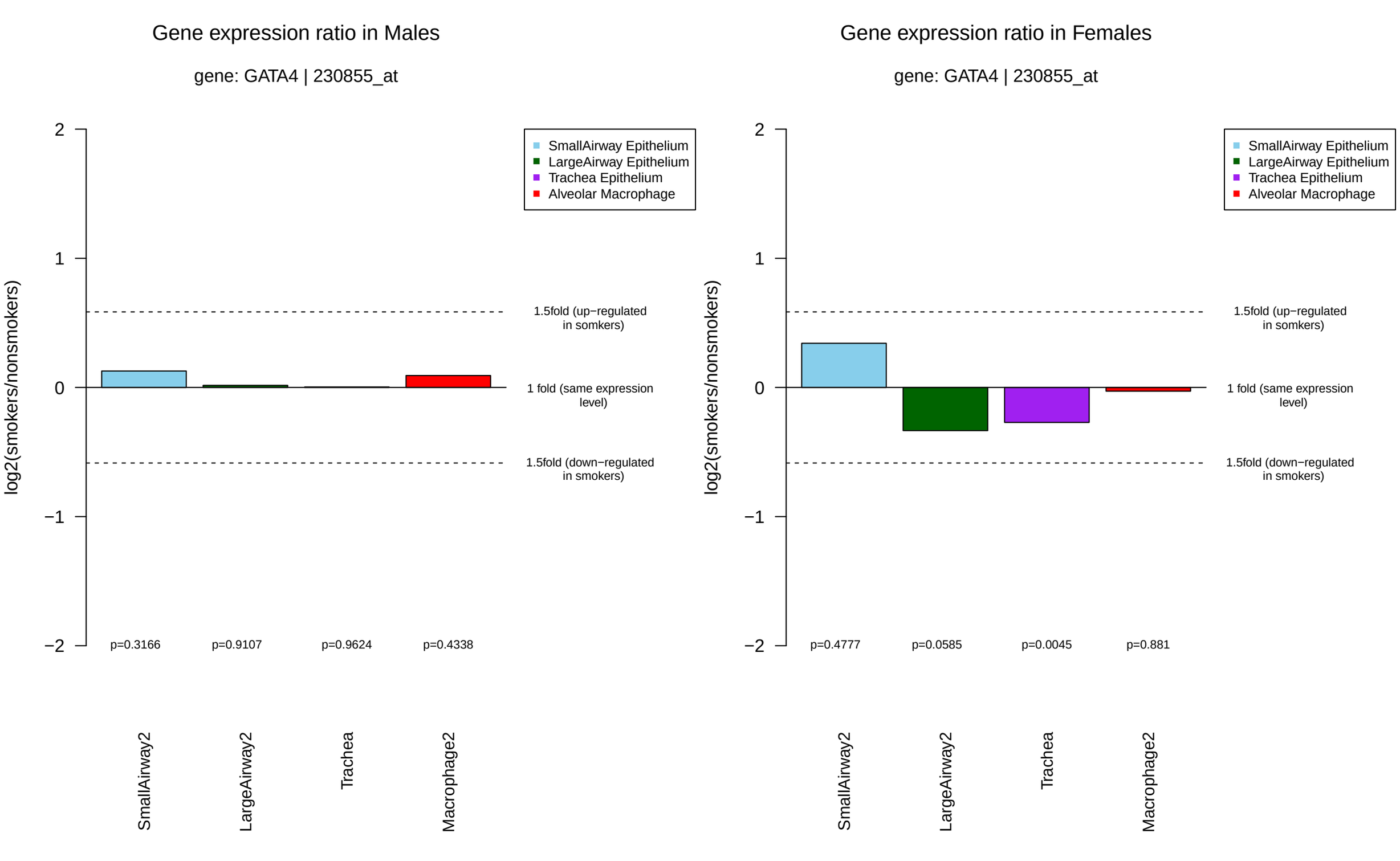
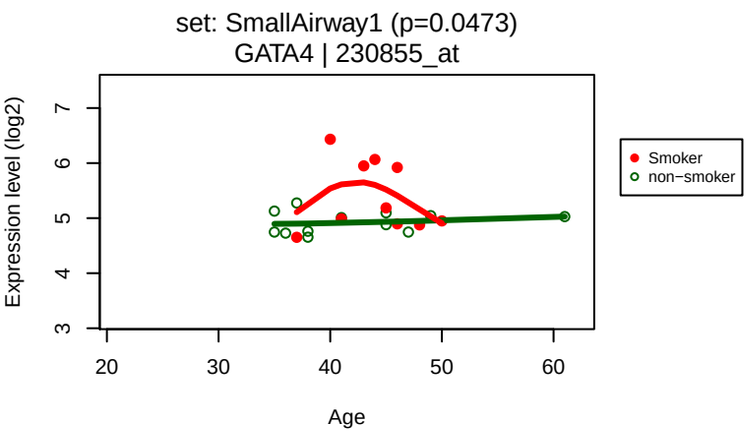
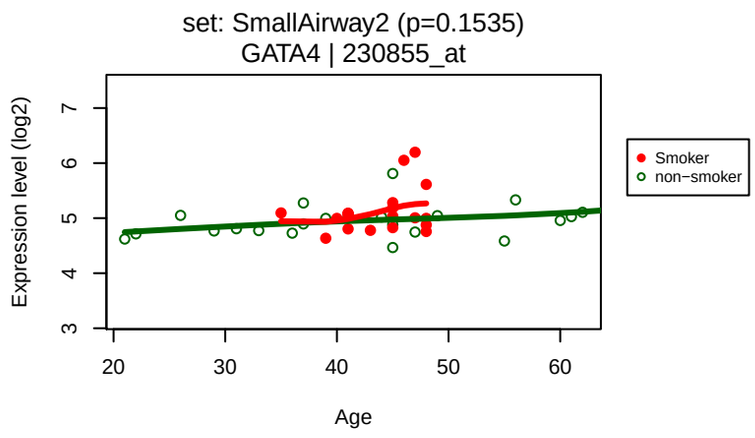
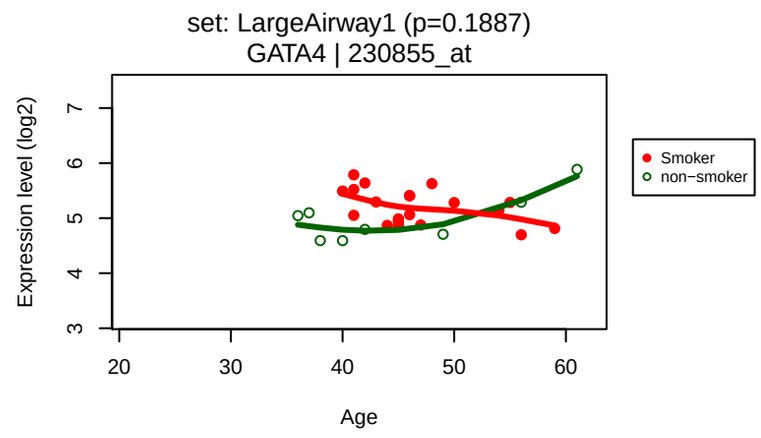

Gene expression of smokers and nonsmokers
gene: GATA4 | 230855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0473
p=0.1535
p=0.0303
p=0.1887
p=0.3467
p=0.3994
p=0.7251
p=0.5636
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

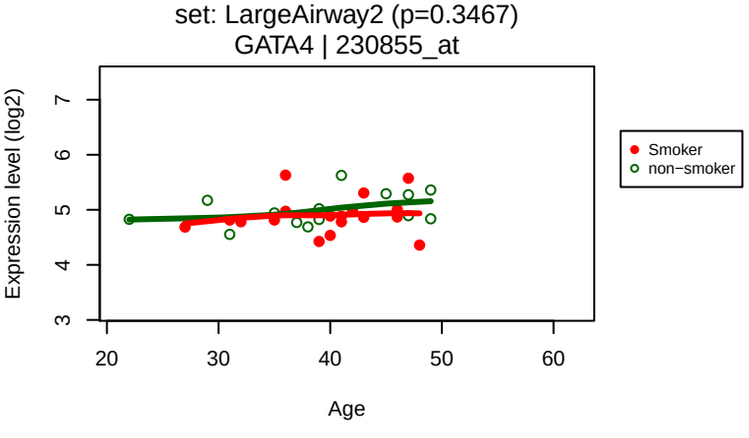

Gene expression ratio (smokers/nonsmokers)
gene: GATA4 | 230855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0473
p=0.1535
p=0.0303
p=0.1887
p=0.3467
p=0.3994
p=0.7251
p=0.5636
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

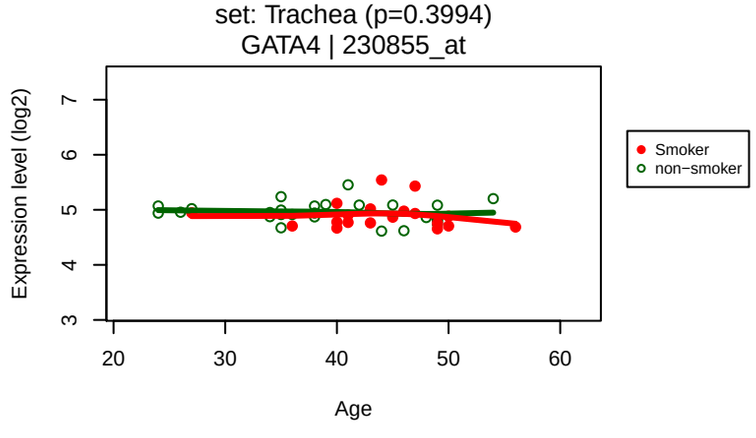

Gene expression ratio in Males
gene: GATA4 | 230855_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3166
p=0.9107
p=0.9624
p=0.4338
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GATA4 | 230855_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4777
p=0.0585
p=0.0045
p=0.881
SmallAirway2
LargeAirway2
Trachea
Macrophage2

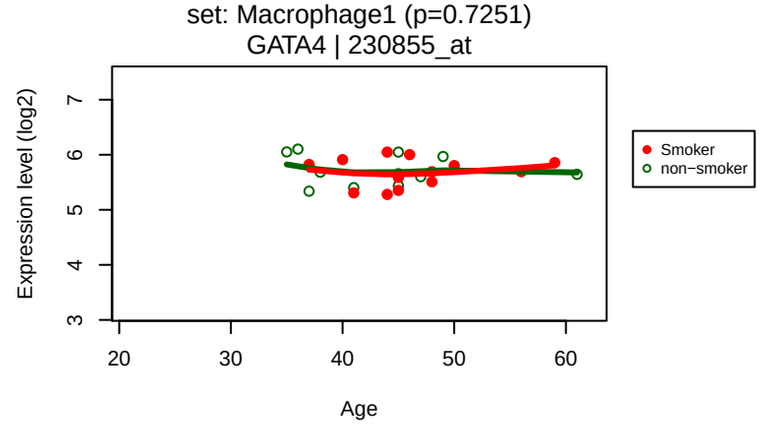

set: SmallAirway1 (p=0.0473)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker

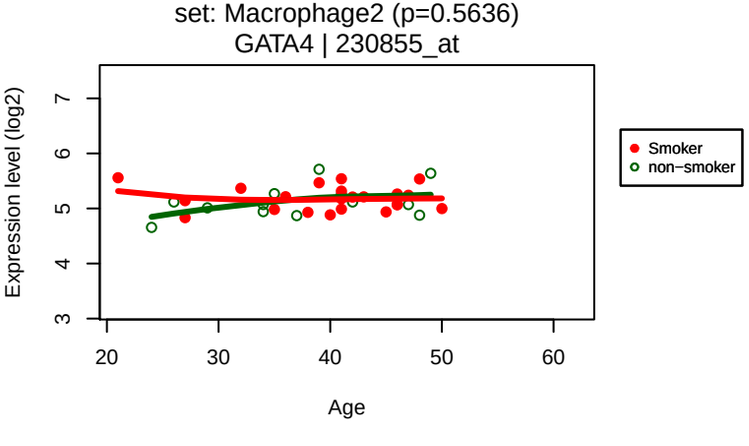

set: SmallAirway2 (p=0.1535)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker

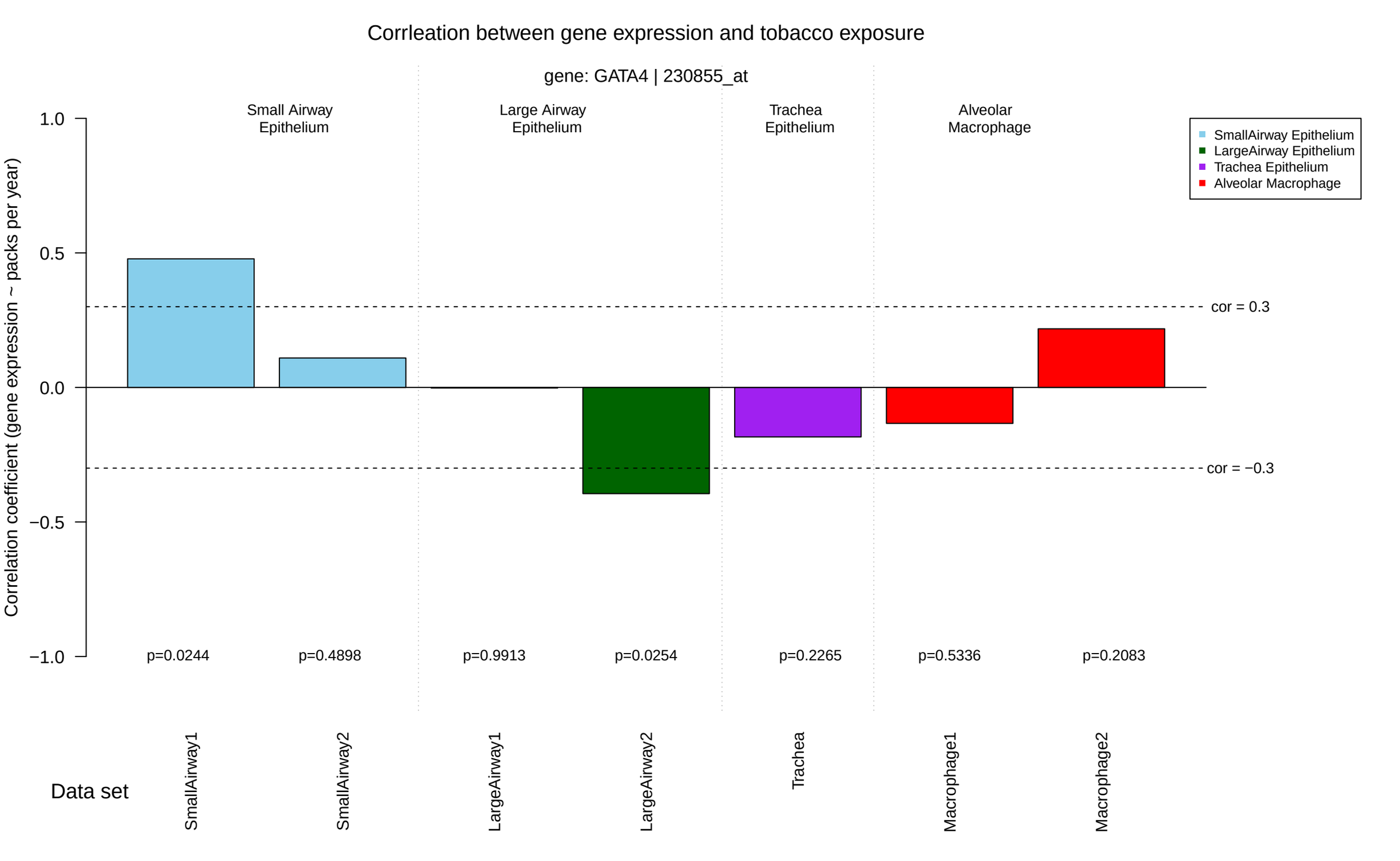

set: LargeAirway1 (p=0.1887)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3467)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3994)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7251)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5636)
GATA4 | 230855_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GATA4 | 230855_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0244
p=0.4898
p=0.9913
p=0.0254
p=0.2265
p=0.5336
p=0.2083
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2